This principle is a resounding affirmation that the Michelson—Morley experiment was correct, and that regardless of how accurate light-measuring devices may become in the future, they must always continue to give the same result: a universal speed of light.

- *The principle of relativity:* Whatever might be their nature, the laws of physics must treat all states of motion on an equal footing.

This principle is a resounding rejection of absolute space: If the laws of physics did not treat all states of motion (for example, that of the Sun and that of the Earth) on an equal footing, then using the laws of physics, physicists would be able to pick out some "preferred" state of motion (for example, the Sun's) and define it as the state of "absolute rest." Absolute space would then have crept back into physics. We shall return to this later in the chapter.

From the absoluteness of the speed of light, Einstein deduced, by an elegant logical argument described in Box 1.1 below, that if you and I move relative to each other, *what I call space must be a mixture of your space and your time, and what you call space must be a mixture of my space and my time.*

This "mixing of space and time" is analogous to the mixing of directions on Earth. Nature offers us two ways to reckon directions, one tied to the Earth's spin, the other tied to its magnetic field. In Pasadena, California, magnetic north (the direction a compass needle points) is offset eastward from true north (the direction toward the Earth's spin axis, that is, toward the "North Pole") by about 20 degrees; see Figure 1.2. This means that in order to travel in the magnetic north direction, one must travel partly (about 80 percent) in the true north direction and partly (about 20 percent) toward true east. In this sense, *magnetic north is a mixture of true north and true east;* similarly, true north is a mixture of magnetic north and magnetic west.

To understand the analogous mixing of space and time *(your space is a mixture of my space and my time, and my space is a mixture of your space and your time),* imagine yourself the owner of a powerful sports car. You like to drive your car down Colorado Boulevard in Pasadena, California, at extremely high speed in the depths of the night, when I a policeman, am